TAMATAM SRIKANTH

gs25mtech11116@iith.ac.in | srikanthtamatam15@gmail.com | +91 7901549257 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.12
Narayan Junior College	Class XII (Board of Intermediate Education Andhra Pradesh, India)	2020	9.73%
Ravindra Vidya Nikethan	Class X (Board of Secondary Education Andhra Pradesh, India)	2018	9.80%

INTERNSHIP

Data Science Intern | Brain O Vision Pvt. Ltd.

Automated Loan Approval Prediction System

- Developed an automated loan approval system by benchmarking Decision Tree, KNN, SVM, and Random Forest on applicant financial features using scikit-learn, achieving 98.01% test accuracy.
- Designed a preprocessing pipeline using LabelEncoder and StandardScaler, and identified CIBIL score, income, and loan amount as top predictors through feature importance analysis.
- Implemented a real-time prediction script using pickle, delivering instant automated loan approval outcomes from user-provided features.

PROJECTS

Deep Learning-Based Spray Analysis using Computer Vision | M.Tech Thesis Project (ongoing)

- Building an end-to-end deep learning framework combining CNN backbones with temporal models ConvLSTM and Transformers to predict spray parameters.
- Designing an automated spray monitoring pipeline using computer vision for real-time parameter estimation and quality assessment from high-speed image sequences.
- Targeting cross-domain applications including automotive fuel injection, agricultural pesticide spraying, and industrial spray systems for data-driven efficiency optimization.

Chest Disease Classification Using Deep Learning & Explainable AI | B.Tech Thesis Project

- Engineered a 6-class chest disease image classification pipeline using TensorFlow/Keras across 21,000+ X-rays, achieving 82% test accuracy with Xception.
- Fine-tuned deep CNNs (Xception, DenseNet121) using Adam and learning rate scheduling (ReduceLROnPlateau), achieving 99% recall and 96% precision on pneumonia detection.
- Integrated Grad-CAM heatmaps to highlight clinically relevant lung regions, enabling visual validation for clinical decision support.

Customer Trends and Sales Analytics using SQL, Python and Power BI

- Built an end-to-end analytics pipeline using SQL, Python (Pandas, NumPy) and Power BI to analyze customer behavior and sales performance.
- Formulated SQL queries using JOINs, CTEs, Window Functions, Aggregate Functions and Subqueries to analyze customer behavior, revenue trends and product performance.
- Designed an interactive Power BI dashboard featuring KPI cards, sales trends, customer demographics and regional insights to support data-driven decisions.

SKILLS

Programming Languages: Python, SQL, C++, C

Data Science & ML: Machine Learning, Deep Learning, Transfer Learning, Feature Engineering, Data Preprocessing, EDA.

Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Matplotlib, Seaborn

Visualization & BI: Power BI, Tableau, Excel, Google Sheets

Database & Tools: MySQL, Git, GitHub, Jupyter Notebook, Google Colab, VS Code, MySQL Workbench

CERTIFICATIONS

- Advanced SQL Certification | 5-Star SQL Rating -- HackerRank
- Python for Data Science, AI & Development -- IBM
- Data Analytics Job Simulation -- Deloitte | Forage
- Career Essentials in Software Development -- Microsoft

RELEVANT COURSES

Machine Learning | Probability & Statistics | Linear Algebra | Optimization Techniques | Database Management Systems | Data Structures & Algorithms | Natural Language Processing | Data Warehousing and Data Mining

POSITIONS OF RESPONSIBILITY

Teaching Assistant | Analysis and Methods for Sustainable Engineering | IIT Hyderabad

- Delivered Python tutorials covering programming fundamentals and scientific computing libraries for optimization and time-series forecasting.
- Assisted students in implementing linear optimization and cost-benefit analysis using Python-based computational tools.

EXTRACURRICULAR

- Participated in the Kurnool District Chess Tournament 2015, organized by the Kurnool District Olympic Association.
- Represented school in the District-Level Handball Championship 2014-15 (U-15).
- Won First Place in Cricket, Kho-Kho, Kabaddi, and Drawing & Painting at school-level co-curricular competitions, earning Certificates of Merit (2016).